Randomized Controlled Trial

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What is a Causal Effect?

A causal effect is defined as a comparison between two states of the world:

(*i*) The actual state of the world

(*ii*) The alternative state of the world (counterfactual state)

Think about a simple example:

- In the actual state of the world, a person takes **aspirin** for a headache and reports the severity of the headache one hour later.
- In the counterfactual state of the world, that same person takes **nothing** for a headache and reports the severity of the headache one hour later.

What is a Causal Effect? Cont'd

Causal effect of the aspirin treatment:

- The causal effect of the aspirin is the difference in the severity of headache between the two states of the world.
- However, counterfactuals are not realized because they are hypothetical states of the world.
- **We have missing data on the counterfactual outcome!**

Treatment Variable

- The treatment allocation is defined through a binary variable

$$D_i = \begin{cases} 1 & \text{if a particular unit } i \text{ receives the treatment.} \\ 0 & \text{if it does not.} \end{cases}$$

- Each unit will have two **potential outcomes**

$$\begin{cases} Y_i^1 & \text{if unit } i \text{ receives the treatment.} \\ Y_i^0 & \text{if it does not (control state).} \end{cases}$$

- Each unit i has an observable or **actual outcome** Y_i, and we have the switching equation:

$$Y_i = D_i Y_i^1 + (1 - D_i) Y_i^0$$

Treatment Effects

The unit-specific treatment effect, or causal effect, is given by

$$\delta_i = Y_i^1 - Y_i^0$$

Three Important Estimators

- ATE: the average treatment effect for the whole population
- ATT: the average treatment effect for the treatment group
- ATU: the average treatment effect for the untreated group (or control group)

Average Treatment Effect (ATE)

$$ATE = E[\delta_i] = E[Y_i^1 - Y_i^0] = E[Y_i^1] - E[Y_i^0]$$

- ATE is the average treatment effect for the whole population.
- Since we cannot observe both the actual and counterfactual worlds simultaneously, **ATE is inherently unknowable.**

$\Rightarrow$ *"Fundamental Problem of Causal Inference"*

Average Treatment Effect for Treatment Group (ATT)

$$
\begin{aligned}
ATT = E[\delta_i | D_i = 1] &= E[Y_i^1 - Y_i^0 | D_i = 1] \\
&= E[Y_i^1 | D_i = 1] - E[Y_i^0 | D_i = 1]
\end{aligned}
$$

- $D_i = 1$ refers to the units that are assigned to the treatment group.
- ATT is the population mean treatment effect for the group of units that had been assigned the treatment.
- ATT will differ from the ATE if D_i correlates with $Y_i^1 - Y_i^0$.
- **ATT is unknown** as well because $Y_i^0 | D_i = 1$ is not observed.

Average Treatment Effect for Untreated Group (ATU)

$$
\begin{aligned}
ATU = E[\delta_i | D_i = 0] &= E[Y_i^1 - Y_i^0 | D_i = 0] \\
&= E[Y_i^1 | D_i = 0] - E[Y_i^0 | D_i = 0]
\end{aligned}
$$

- $D_i = 0$ refers to the units that are assigned to the control group.
- ATU is the population mean treatment effect for those units sorted into the control group.
- Typically in an observational setting $ATT \neq ATU$.
- **ATU is unknown** because $Y_i^1 | D_i = 0$ is not observed.

Example

Naïve experiment: *"What is the effect of a college degree on income?"*

- Treatment variable: $D_i = \begin{cases} 1 & \text{if person } i \text{ has a college degree} \\ 0 & \text{if it does not} \end{cases}$
- Observed outcome Y_i: Annual income in USD
- Mean observed outcomes (same number of people in control and treatment group):

	No college degree ($D = 0$)	College degree ($D = 1$)
Y_0	\$50,000	unobservable
Y_1	unobservable	\$80,000

Can we conclude that since, on average, college degree holders earn 30,000 USD more, we should encourage more people to get a college degree?

Example (Cont'd)

Let us now assume we could turn back time and observe the respective counterfactuals for our test subjects:

	No college degree ($D = 0$)	College degree ($D = 1$)
Y_0	\$50,000	\$60,000
Y_1	\$55,000	\$80,000

This lets us compute:

- $ATT = \$80,000 - \$60,000 = \$20,000$
- $ATU = \$55,000 - \$50,000 = \$5,000$
- $ATE = \frac{\$55,000 + \$80,000}{2} - \frac{\$50,000 + \$60,000}{2} = \$12,500$

Example (Cont'd)

We can interpret the results as follows:

- *ATT*: On average, the treatment group's income increases by \$20,000 by getting a college degree.
- *ATU*: On average, the control group's income increases by \$5,000 by getting a college degree.
- *ATE*: On average, getting a college degree increases the income by \$12,500.

What can we say about our experimental design if the values of *ATT* and *ATU* are so far apart?

Example (Cont'd)

Let's go back to the real world:

	No college degree ($D = 0$)	College degree ($D = 1$)
Y_0	\$50,000	unobservable
Y_1	unobservable	\$80,000

Due to unobservable data, the only estimator we can directly compute is the **Simple Difference in Outcomes (SDO)**, which is defined as:

$$SDO = E[Y^1|D = 1] - E[Y^0|D = 0]$$

In this case, we have:

$$SDO = \$80,000 - \$50,000 = \$30,000$$

Example (Cont'd)

While the real *ATE* is \$12,500, the naïve estimator *SDO* yields a result of \$30,000.

"Why is there such a big difference?"

Possible reasons:

- People who choose to go to college may differ systematically from those who do not (e.g., motivation, family background).
- Individuals with higher innate ability or cognitive skills are both more likely to attend college and to earn higher wages.
- Factors such as networking opportunities, regional labor markets, or parental education affect both the likelihood of attending college and income outcomes.

Where does the bias come from?

The *SDO* can be decomposed into three parts:

$$
\begin{aligned}
SDO &= E[Y^1|D=1] - E[Y^0|D=0] \\
&= ATE + E[Y^0|D=1] - E[Y^0|D=0] + (1-\pi)(ATT - ATU),
\end{aligned}
$$

where π is the share of people who received the treatment and $1-\pi$ is the share of people in the control group.

Where does the bias come from? (Cont'd)

To understand where these parts on the right-hand side originate, we need to decompose the parameter of interest, *ATE*, into its basic building blocks. *ATE* is equal to the weighted sum of conditional average expectations, *ATT* and *ATU*:

$$
\begin{aligned}
ATE &= \pi ATT + (1-\pi)ATU \\
&= \pi E[Y^1 \mid D=1] - \pi E[Y^0 \mid D=1] \\
&\quad + (1-\pi)E[Y^1 \mid D=0] - (1-\pi)E[Y^0 \mid D=0] \\
&= \Big\{\pi E[Y^1 \mid D=1] + (1-\pi)E[Y^1 \mid D=0]\Big\} \\
&\quad - \Big\{\pi E[Y^0 \mid D=1] + (1-\pi)E[Y^0 \mid D=0]\Big\}
\end{aligned}
$$

where π is the share of people who receive treatment and $1-\pi$ is the share of people in the control group.

Where does the bias come from? (Cont'd)

Because the conditional expectation notation is a little cumbersome, let's introduce the following:

$$
\begin{aligned}
E[Y^1 \mid D = 1] &= a \\
E[Y^1 \mid D = 0] &= b \\
E[Y^0 \mid D = 1] &= c \\
E[Y^0 \mid D = 0] &= d \\
ATE &= e
\end{aligned}
$$

	$D = 0$	$D = 1$
Y_0	d	c
Y_1	b	a

Where does the bias come from? (Cont'd)

Now we can write the ATE as follows:

$$\begin{aligned}
e &= \{\pi a + (1-\pi)b\} - \{\pi c + (1-\pi)d\} \\
e &= \pi a + b - \pi b - \pi c - d + \pi d \\
e &= \pi a + b - \pi b - \pi c - d + \pi d + (a-a) + (c-c) + (d-d) \\
0 &= e - \pi a - b + \pi b + \pi c + d - \pi d - a + a - c + c - d + d \\
a - d &= e - \pi a - b + \pi b + \pi c + d - \pi d + a - c + c - d \\
a - d &= e + (c-d) + a - \pi a - b + \pi b - c + \pi c + d - \pi d \\
a - d &= e + (c-d) + (1-\pi)a - (1-\pi)b + (1-\pi)d - (1-\pi)c \\
a - d &= e + (c-d) + (1-\pi)(a - c - b + d)
\end{aligned}$$

Where does the bias come from? (Cont'd)

Substituting the initial expressions yields:

$$
\begin{aligned}
& E[Y^1 \mid D = 1] - E[Y^0 \mid D = 0] \\
& = ATE + \Big(E[Y^0 \mid D = 1] - E[Y^0 \mid D = 0] \Big) + (1 - \pi)(ATT - ATU),
\end{aligned}
$$

which is the *SDO* decomposition used above.

Where does the bias come from? (Cont'd)

To be more precise, we have:

$$\underbrace{\frac{1}{N_T}\sum_{i=1}^{n}(y_i|d_i=1)-\frac{1}{N_U}\sum_{i=1}^{n}(y_i|d_i=0)}_{SDO} = \underbrace{E[Y^1]-E[Y^0]}_{\text{Average Treatment Effect}}$$

$$+\underbrace{E[Y^0|D=1]-E[Y^0|D=0]}_{\text{Selection Bias}}$$

$$+\underbrace{(1-\pi)(ATT-ATU)}_{\text{Heterogeneous Treatment Effect Bias}}$$

Thus, the bias comes from the **Selection Bias** and the **Heterogeneous Treatment Effect Bias**.

Understanding the biases

So, we have: **SDO = ATE + Selection Bias + Heterogeneous Treatment Effect Bias**

The three components are:

1. Average Treatment Effect (ATE): The parameter of interest
2. Selection bias: A description of the differences between the two groups if there had never been a treatment
3. Heterogeneous Treatment Effect Bias: Bias of different treatment effects multiplied by the share of the control group

Understanding the biases (Cont'd)

Let's verify the decomposition using our example:

	No college degree ($D = 0$)	College degree ($D = 1$)
Y_0	\$50,000	\$60,000
Y_1	\$55,000	\$80,000

- ATE: $\frac{\$55{,}000+\$80{,}000}{2} - \frac{\$50{,}000+\$60{,}000}{2} = \$12,500$
- Selection Bias: $\$60,000 - \$50,000 = \$10,000$
- Heterogeneous Treatment Effect Bias: $(1 - \frac{1}{2}) \cdot (\$20,000 - \$5,000) = \$7,500$
- SDO: $\$30,000$ ($= \$12,500 + \$10,000 + \$7,500$ ✓)

Can we eliminate the biases?

- Unfortunately, the two biases cannot be calculated as they depend on unobservable counterfactuals.
- However, we can introduce the so-called **Independence Assumption** to deal with the biases which requires the groups to be assigned independently.
- In our example, we looked at a highly biased allocation to the control and treatment group (motivation, family background, ...). Would it be possible to run **this** experiment by assigning people independently of their potential outcomes?

Independence Assumption

If we can ensure that the control and treatment group were assigned independently, we have:

- $E[Y^1|D=1] - E[Y^1|D=0] = 0$
- $E[Y^0|D=1] - E[Y^0|D=0] = 0$

This means, the expected potential outcome (Y^1 or Y^0) is the same for the two groups.

Independence Assumption

Successfully conducting such a **Randomized Controlled Trial (RCT)** eliminates both the Selection Bias and the Heterogeneous Treatment Effect Bias:

- $ATT = E[Y^1|D = 1] - E[Y^0|D = 1]$
- $ATU = E[Y^1|D = 0] - E[Y^0|D = 0]$
- Heterogeneous Treatment Effect: $(1 - \pi)(ATT - ATU) = 0$.

This means that if the treatment is independent of the potential outcomes, we have $SDO = ATE$ which means we can **"observe the unobservable"**!

SUTVA

Besides the Independence Assumption, we also need the **Stable Unit Treatment Value Assumption (SUTVA)**:

- Each treated unit receives the same dose.
- No spillovers ("externalities"): The potential outcome observation on one unit should be unaffected by the particular assignment of treatments to the other units.
- No general equilibrium effects.

Statistical significance

Statistical significance of an observed effect is usually determined by the corresponding p-value, which is defined as:

"The probability of obtaining test results at least as extreme as the result actually observed, under the assumption that the null hypothesis is correct."

In other words, the p-value is the probability to which the data supports the null hypothesis (no treatment effect).

The p-value is a piece of evidence, not a definitive proof!

Statistical significance (Cont'd)

The p-value is **not**...

- ...the probability that the null hypothesis is true.
- ...the probability that the alternative hypothesis is false.
- ...automatically indicating a significant result if it is below 0.05. This significance level is a mere convention and is not a strict boundary.
- ...indicating the size or importance of the observed effect.

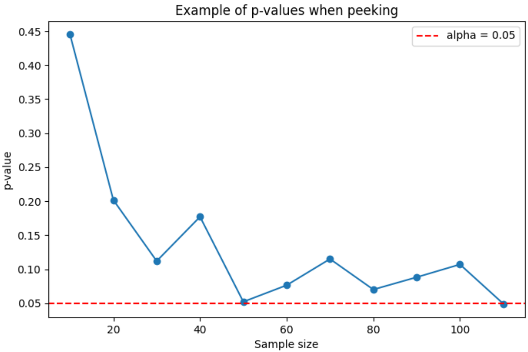
Statistical significance (Cont'd)

Alternatives to using p-values:

- Confidence Intervals:
  - Providing more detail about effect sizes.
  - e.g., *"The intervention reduced blood pressure by -5.2 mmHg (95% CI: -8.3 to -2.1)."*
- Bayesian Inference:
  - Method: Starting from a prior belief, calculating a posterior distribution of the effect size.
  - Offers intuitive insights for decision makers.
  - Calculating a Bayes Factor tells us how much more evidence one hypothesis provides compared to another.

Fixed vs. Adaptive Sample Size Testing

- As traditional RCTs are expensive and lengthy, many RCTs on online platforms are nowadays conducted without a predetermined sample size.
- What might be the problem of stopping the experiment at the first time we hit e.g. $p < 0.05$?



- Every time you peek, you start a new hypothesis test
- Selection bias (real p is much higher!)

Selecting the sample size

Before conducting an RCT, the sample size should be determined. The required number to reach statistically significant results can be **roughly estimated** by taking into account:

- Significance level
- Marginal error (Relative Confidence Interval)
- Power (Probability of correctly rejecting the null hypothesis when the effect is real)
- Population size

However, this does not guarantee that the experiment will yield statistically significant results!

Guidelines for conducting RCTs

In conclusion, to run an RCT, we need the following steps:

1. Set up the experiment and define the research question (incl. outcome that will be measured, hypotheses, and identification of the population).
2. Make sure that the Independence Assumption and SUTVA hold.
3. Determine the sample size.
4. Conduct the experiment (considering independent, random group allocation) and collect data.
5. Analyze, interpret, and report the results (compute the p-value and potential additional metrics).

Practical example of an RCT

- **Research question:** Is there a causal impact on school assignment submission times when framing grading guidelines as bonus vs. penalty schemes?
- Hypotheses:
  - H_0: There is no significant difference in submission times.
  - H_1: There is a significant difference in submission times.
- Sample size: 135 (68+67)
- Timeline of experiment: 21 Aug 2025 - 02 Sept 2025

Practical example of an RCT (Cont'd)

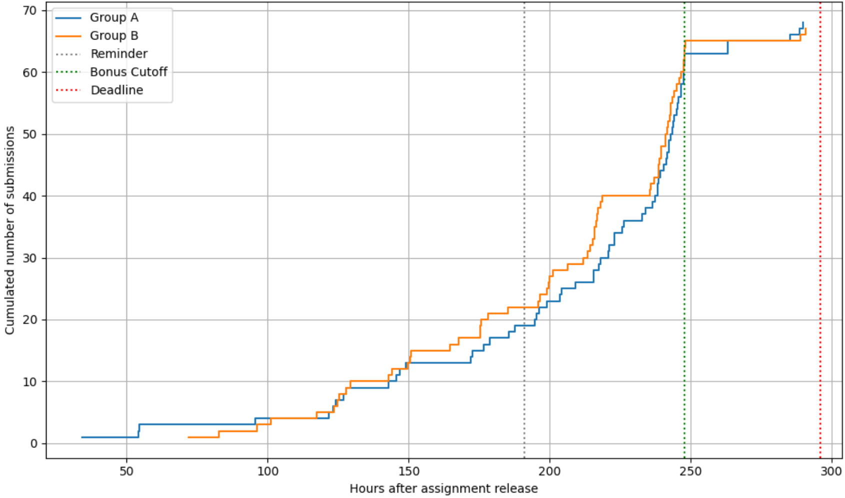
Control Group (A): Bonus Group ("Homework 1"):

This homework is worth 50 marks, with the opportunity to earn 10 bonus marks. It is due on **September 2, 2025, 23:59**. Late submissions will not be accepted. Please submit your answers as a single PDF file through Canvas. If you submit by **August 31, 2025, 23:59**, you will receive 10 bonus marks.

Treatment Group (B): Penalty Group ("HW 1"):

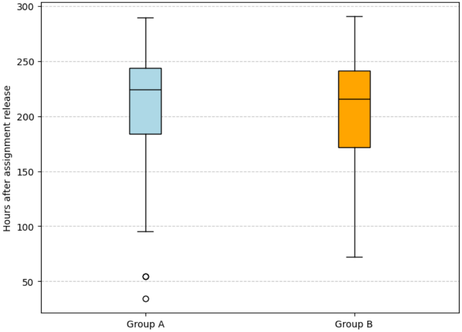
This homework is worth 50 marks, and you have been awarded 10 bonus marks in advance. It is due on **September 2, 2025, 23:59**. Late submissions will not be accepted. Please submit your answers as a single PDF file through Canvas. If you submit after **August 31, 2025, 23:59**, the 10 bonus marks will be deducted.

Practical example of an RCT (Cont'd)



Practical example of an RCT (Cont'd)

Descriptive Statistics:



	Group A	Group B
Mean	206.83	201.12
Median	224.40	215.95
St. Dev.	55.38	50.43

Practical example of an RCT (Cont'd)

Is the overall difference in submission duration statistically significant?

- T-statistic: 0.6272
- P-value: 0.53163

⇒ **No statistical significance**

What if we only look at the period between the Reminder and the Bonus Cutoff?

- T-statistic: 0.6853
- P-value: 0.49508

⇒ **Still no statistical significance**

Similar results for other approaches such as Linear Regression, Cox Proportional Hazards Model, Mann-Whitney U test, ...

Practical example of an RCT (Cont'd)

What could have been done better?

- Larger sample size: The effect might be small which requires a larger sample size to detect.
- Balance check: Confirm that the randomization worked as expected by comparing basic covariates (Age, gender, prior grades, etc.) between groups.
- Mitigation of network effects to ensure SUTVA holds.
- Highlighting bonus/penalty more distinctly and ensuring the students do not skim the instructions.
- Collecting more data (e.g., quality of submissions or time spent on assignment).

$\Rightarrow$ "No significance" does **not** mean "no effect". Careful experimental design might lead to insights not captured in the current analysis.